Object Oriented Analysis and Design: Case Study 1

HEART HEALTH IMAGING AND RECORDING SYSTEM

A. List of objects you have identified:

Entity: Represents data and business objects within the system, encapsulating core data and related behavior.

- User: Aggregates all common system user information and implements role-based access control.
- Role: Represents different user roles within the system, enabling role-based separation of business logic.
- CTTest: Encapsulates all data points required for a CT scan, including a reference to the results file and a link to the patient ID.
- Appointment: Handles the scheduling of CT scan appointments for patients.
- PatientRecord: Represents the patient's information file, including personal details and health history, linked to the unique patient ID.

Control: Manages the flow of data between boundary and entity objects, implementing the logic required to fulfill the system's use cases.

- Login: Handles user authentication and access control.

Boundary: Represents interactions between the system and its actors, encapsulating the interface and interaction logic.

- Patient: Contains business logic specific to patients, such as viewing CT scan results.
- Receptionist: Manages patient interactions and schedules CT scan appointments, including acquiring patient information and generating unique patient IDs.
- CTScanTechnician: Performs CT scans and records results to the patient's record.
- HeartSpecialist (Doctor): CT scan results, determine the patient's risk, and communicate the findings to the patient.

B. CRC diagram for each object identified above

Class:	User
Description:	Represents the data structure for a system user.
Responsibility:	Collaborator:
Store common user information	Role
Implement roles for access	Role

Class:	Role
Description:	Represents different user roles within the system, enabling role-based separation of business logic.
Responsibility:	Collaborator:
Abstract user roles and permissions	User
Enable role-based separation of business logic	User

Class:	CTTest
Description:	Encapsulates all data points required for a CT scan, including a reference to the results file and a link to the patient ID.
Responsibility:	Collaborator:
Store CT scan data points	User, Patient, HeartSpecialist
Reference to the results file	Patient, HeartSpecialist
Link to the patient ID	Patient, HeartSpecialist

Class:	Appointment
Description:	Handles the scheduling of CT scan appointments for patients.
Responsibility:	Collaborator:
Schedule CT scans	Receptionist
Provide appointment information	CTScanTechnician

Class:	Patient
Description:	Contains business logic specific to patients, such as viewing CT scan results.
Responsibility:	Collaborator:
View CT scan results	CTTest
Receive error message if data not available	CTTest

Class:	Receptionist
Description:	Manages patient interactions and schedules CT scan appointments, including acquiring

	patient information and generating unique patient IDs.
Responsibility:	Collaborator:
Acquire patient information	Appointment
Schedule CT exams	Appointment
Generate unique 5-digit patient ID	User

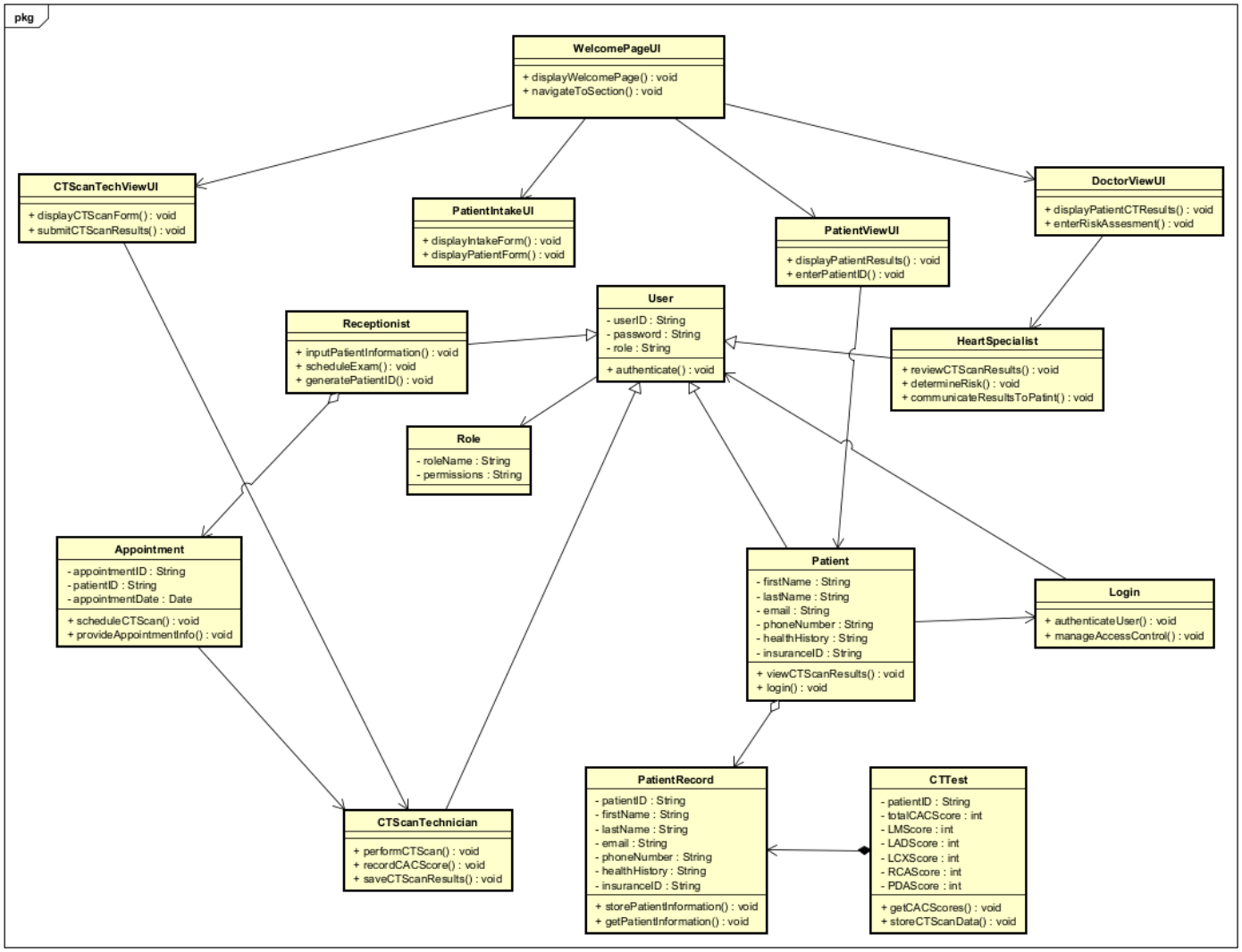
Class:	CTScanTechnician
Description:	Performs CT scans and records results to the patient's record.
Responsibility:	Collaborator:
Acquire patient information	Appointment
Record results to patient record	CTTest

Class:	HeartSpecialist (Doctor)
Description:	Reviews CT scan results, determines the patient's risk, and communicates the findings to the patient.
Responsibility:	Collaborator:
View patient CT data	CTTest
Initiate risk evaluation program	CTTest
Communicate with patients about next steps	Patient

Class:	Login
Description:	Handles user authentication and access control.
Responsibility:	Collaborator:
Authenticate user	User
Manage access control	User

Class:	PatientRecord
Description:	Represents the patient's information file, including personal details and health history, linked to the unique patient ID.
Responsibility:	Collaborator:
Store patient information	Receptionist
Link to unique patient ID	Receptionist

C. Class Diagram


pkg
WelcomePageUI
+ displayWelcomePage() : void
+ navigateToSection() : void
CTScanTechViewUI
+ displayCTScanForm() : void
+ submitCTScanResults() : void
PatientIntakeUI
+ displayIntakeForm() : void
+ displayPatientForm() : void
PatientViewUI
+ displayPatientResults() : void
+ enterPatientID() : void
DoctorViewUI
+ displayPatientCTResults() : void
+ enterRiskAssesment() : void
User
- userID : String
- password : String
- role : String
+ authenticate() : void
Receptionist
+ inputPatientInformation() : void
+ scheduleExam() : void
+ generatePatientID() : void
HeartSpecialist
+ reviewCTScanResults() : void
+ determineRisk() : void
+ communicateResultsToPatint() : void
Role
- roleName : String
- permissions : String
Appointment
- appointmentID : String
- patientID : String
- appointmentDate : Date
+ scheduleCTScan() : void
+ provideAppointmentInfo() : void
Patient
- firstName : String
- lastName : String
- email : String
- phoneNumber : String
- healthHistory : String
- insuranceID : String
+ viewCTScanResults() : void
+ login() : void
Login
+ authenticateUser() : void
+ manageAccessControl() : void
PatientRecord
- patientID : String
- firstName : String
- lastName : String
- email : String
- phoneNumber : String
- healthHistory : String
- insuranceID : String
+ storePatientInformation() : void
+ getPatientInformation() : void
CTTest
- patientID : String
- totalCACScore : int
- LMScore : int
- LADScore : int
- LCXScore : int
- RCAScore : int
- PDAScore : int
+ getCACScores() : void
+ storeCTScanData() : void
CTScanTechnician
+ performCTScan() : void
+ recordCACScore() : void
+ saveCTScanResults() : void